

Introduction to NLP

Definition

Natural Language Processing (NLP) is a branch of computer science and artificial intelligence that focuses on the interaction between computers and human language.

Aim

NLP aims to enable computers to understand, interpret, and generate human language in a way that is both meaningful and useful.

Technical Goal

NLP empowers machines to process and analyze vast amounts of textual data, unlocking valuable insights and enabling efficient communication.

Challenges

NLP faces challenges like ambiguity in language, variations in dialects and accents, and the need for robust algorithms to handle complex linguistic structures.

The Evolution of NLP: From Rule-Based Systems to Deep Learning and Transformers

1

Rule-Based Systems (1950s-1980s)

Early NLP systems relied on handcrafted rules and linguistic patterns to process language, requiring extensive knowledge of grammar and syntax.

2

Statistical NLP (1990s-2010s)

The rise of machine learning brought about statistical NLP, which leverages statistical models trained on large datasets to identify patterns and make predictions about language.

3

Deep Learning Era (2010s-Present)

Deep learning, particularly recurrent neural networks (RNNs) and transformers, has revolutionized NLP, enabling models to learn complex language representations and achieve state-of-the-art performance on various tasks.

4

Transformers and Pretrained Models (2018-Present)

Introduction of Transformer architecture by Vaswani et al. (2017).

Key Models: BERT (Bidirectional Encoder Representations from Transformers), GPT (Generative Pre-trained Transformer), T5, RoBERTa.

Features: Self-attention mechanism, scalability, and pretraining on massive datasets.

Impact: Significant improvements in a wide range of NLP tasks, state-of-the-art performance.

Unlocking Meaning: Core Concepts in NLP

At its core, NLP centers around breaking down human language into its fundamental components and understanding the relationships between them. Key concepts include:

Text Processing

The foundation of NLP involves cleaning and preparing raw text data, such as removing punctuation, converting to lowercase, and handling special characters, to ensure consistency and improve model performance.

Tokenization

The process of dividing text into individual words or tokens, which serve as the building blocks for further analysis. Tokenization can be as simple as splitting text by spaces or involve more complex rules for handling contractions, punctuation, and special characters.

Stemming & Lemmatization

Techniques for reducing words to their base or root form, such as "running" to "run" or "better" to "good," which helps to consolidate words with similar meanings and reduce vocabulary size, improving model efficiency.

Introduction to NLP. What can be done with NLP

Text summarization

Named entities recognition

Part-of-Speech Tagging
Semantic dependencies

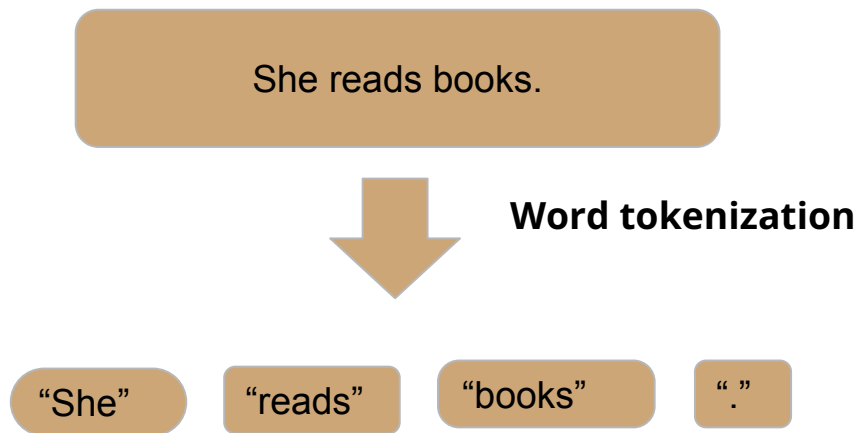
Sentiment Analysis

Text classification

Text Generation

Foundational Principles of NLP - I

Tokenization-first step: splitting text into smaller chunks, typically words or subwords.



Tokenization in Natural Language Processing

Tokenization transforms text into numerical data, crucial for machine learning models in NLP.

The process includes several stages:

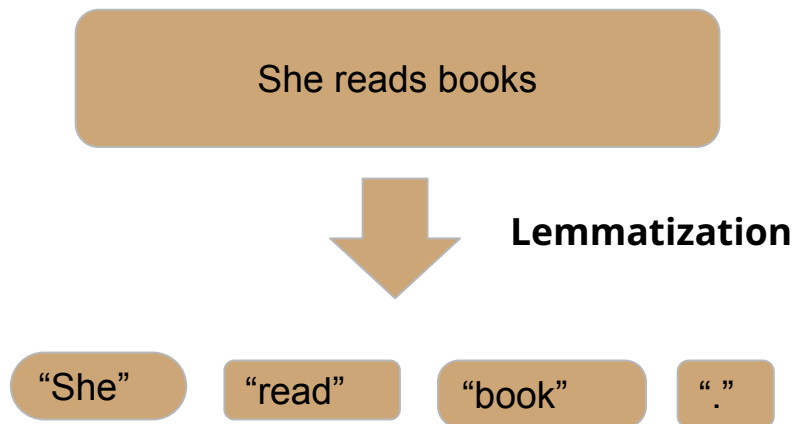
- **Normalization:** Ensures text consistency by cleaning it, like converting letters to lowercase, removing punctuation, or altering accented characters based on the task specifics. While more normalization simplifies data, excessive cleaning can lose context.
- **Pre-tokenization:** Splits the text into segments, such as words, especially when spaces separate them. However, this varies by language.
- **Token Cleanup:** Involves cleaning steps after initial splitting. This might include removing stopwords like "a" or "the," as they add little value to the task at hand.

Balance of Complexity and Context

- **Simple Models:** Require extensive data cleaning to reduce complexity, especially with small datasets.
- **Advanced AI Systems:** Tools like OpenAI's ChatGPT and Google's Gemini retain complexity for nuanced understanding, often leveraging subword tokenization for detailed language analysis.
- **Key Takeaway:** Balance simplicity with essential information to optimize machine learning model training.

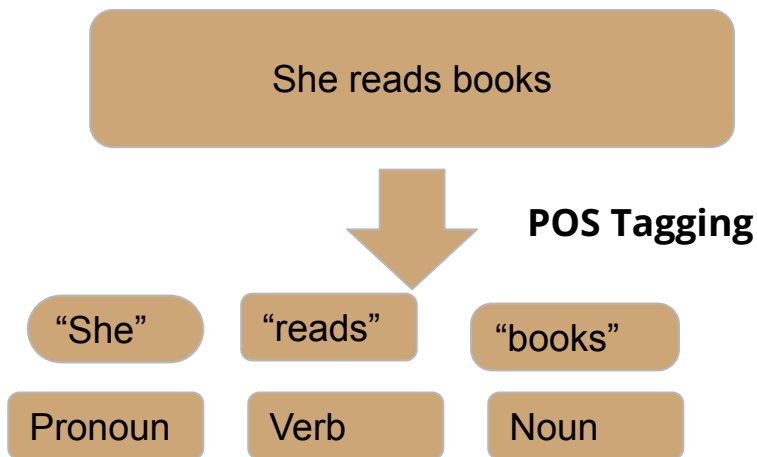
Foundational Principles of NLP - I

Lemmatization: it is a process to change the word to its dictionary form, known as the “lemma”.



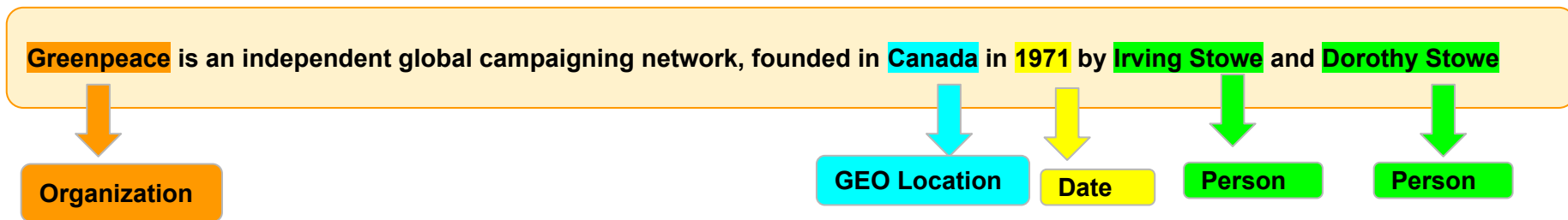
Foundational Principles of NLP - I

Part-of-Speech (POS) Tagging : it is a process of labeling each word (token) in a text with its corresponding part of speech



Foundational Principles of NLP - II

Named Entity Recognition: information extraction from the text that classifies named entities into predefined categories such as the person names, organizations, locations, numbers, specific terms (medical, legal), percentages, etc.



Importance:

Information Retrieval: Helps in improving the accuracy of search systems by focusing on key terms.

Question Answering: Enables systems to answer questions related to specific entities.

Content Recommendation: Helps in content personalization and recommendation by focusing on key entities.

Research: Useful in extracting structured information from massive datasets for academic or corporate research.

Foundational Principles of NLP

- II

Sentiment Analysis is a task to determine and extract the sentiment or emotion in the text. The primary goal is to detect positive/negative/neutral tone of text.

Types:

- **Binary Sentiment Analysis:** Categorizes sentiments as positive or negative.
- **Fine-grained Sentiment Analysis:** Goes beyond binary and might include categories like very positive, positive, neutral, negative, and very negative.
- **Emotion detection:** Determines specific emotions being expressed, e.g., happiness, anger, sadness, etc.

NLP tasks overview

Classification



- Sentiment extraction
- Intent detection
- Language detection
- Topic modeling

Token-level / "Multi"-Classification



- Part-of-speech tagging
- Named entity recognition (NER)
- Dependency parsing
- Constituency parsing

Generation

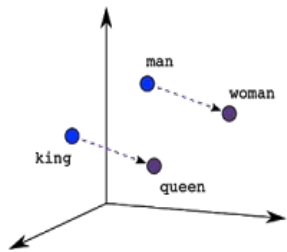


- Machine translation
- Question answering
- Summarization
- Free-form text generation (e.g. chatbots, assistants)

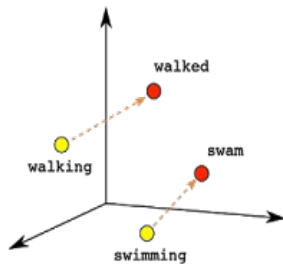
What are Vector Embeddings? Understanding the Foundation of Data Representation.

- Embeddings are representations of data as points in space
- Any type of data can be embedded: text, images, videos, users, music
- The key: locations in this space are semantically meaningful
- Think of it as converting data into coordinates with meaning

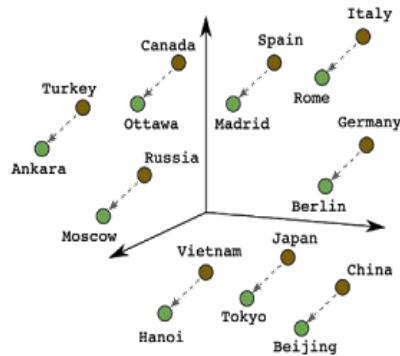
What are Vector Embeddings? Understanding the Foundation of Data Representation



Male-Female



Verb Tense



Country-Capital

Embeddings are a way of representing data—almost any kind of data, like text, images, videos, users, music, whatever—as points in space where the locations of those points in space are semantically meaningful.



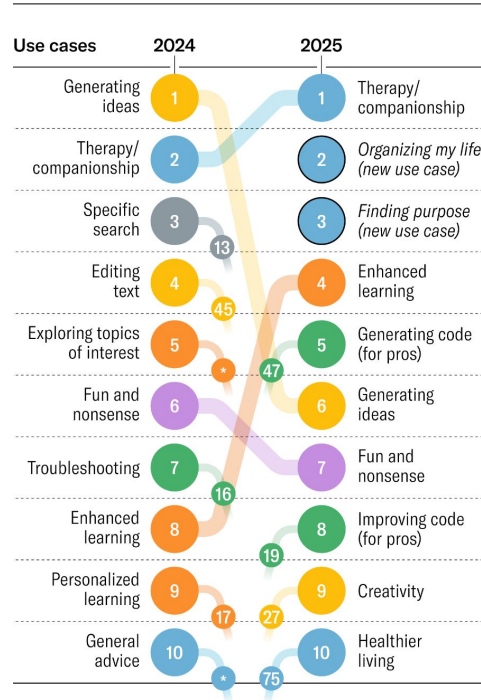
Andrej Karpathy ✓

@karpathy

The hottest new programming language is English

12:14 PM · Jan 24, 2023 · **3.8M** Views

Top 10 Gen AI Use Cases



Top 10 Gen AI Use Cases

The top 10 gen AI use cases in 2025 indicate a shift from technical to emotional applications, and in particular, growth in areas such as therapy, personal productivity, and personal development.

Themes

PERSONAL AND PROFESSIONAL SUPPORT	TECHNICAL ASSISTANCE AND TROUBLESHOOTING
CONTENT CREATION AND EDITING	CREATIVITY AND RECREATION
LEARNING AND EDUCATION	RESEARCH, ANALYSIS, AND DECISION-MAKING

Harvard Business Review

<https://hbr.org/2025/04/how-people-are-really-using-gen-ai-in-2025>

What exactly is an Large Language Models?

Simple Analogy:

Think of an LLM as a super-smart autocomplete on your phone. It's been fed massive amounts of text and code, and it's learned to predict what might come next.

More Technical:

A "large language model" is essentially a type of computer program adept at understanding and generating human language. It's trained on vast datasets consisting of text and code, enabling it to predict and generate coherent language based on the input it receives.

Translate English to Spanish:
What's the time?

What is the expected
weather in Puerto Rico in
December?

Summarize in 15 words
"During extra time, Messi
then scored again to give
Argentina a 3-2 lead.
However, Mbappé scored
another penalty to tie the
game 3-3 with only minutes
remaining, becoming the
second man to score a hat-
trick in a World Cup final.
Argentina then won the
ensuing penalty shoot-out
4-2 to win their third World
Cup."

Generate a tagline for a fruit
juice brand that focused on
sustainable and organic
farming.

LLM

Qué hora es?

In December, Puerto Rico
experiences pleasant
tropical weather with an
average high temperature of
84°F (29°C) and an average
low temperature of around
72°F (22°C).

Mbappé scores a hat-trick
but Argentina wins World
Cup in a penalty shoot-out.

Purely natural, sustainably
sourced. Taste the goodness
of organic fruits with every
sip.

Large Language Models

A Large Language Model (LLM) is a machine learning model designed to handle and generate human-like text based on vast amounts of training data and trained. Foundation Models

LLMs in Action: Cool Things They Can Do

Natural Language Understanding

LLMs excel at understanding and interpreting human language. They can comprehend context, infer meaning, and provide relevant responses to queries.

1

2

Text Generation

These models are capable of generating coherent and contextually relevant text. From writing articles to composing poetry, LLMs can produce content in various styles and genres.

Language Translation

LLMs facilitate seamless language translation by converting text from one language to another while preserving context and meaning.

3

4

Code Generation

With their understanding of programming languages, LLMs can assist developers by generating code snippets, automating repetitive tasks, and offering solutions to coding challenges.

Creative Collaboration

LLMs can serve as collaborative partners in brainstorming sessions, suggesting ideas, refining concepts, and providing inspiration across diverse fields such as art, design, and storytelling

5

6

Overall: Increase productivity!

LLMs can help boost productivity by automating tasks, generating content, and providing valuable insights.

“Attention is All You Need.”

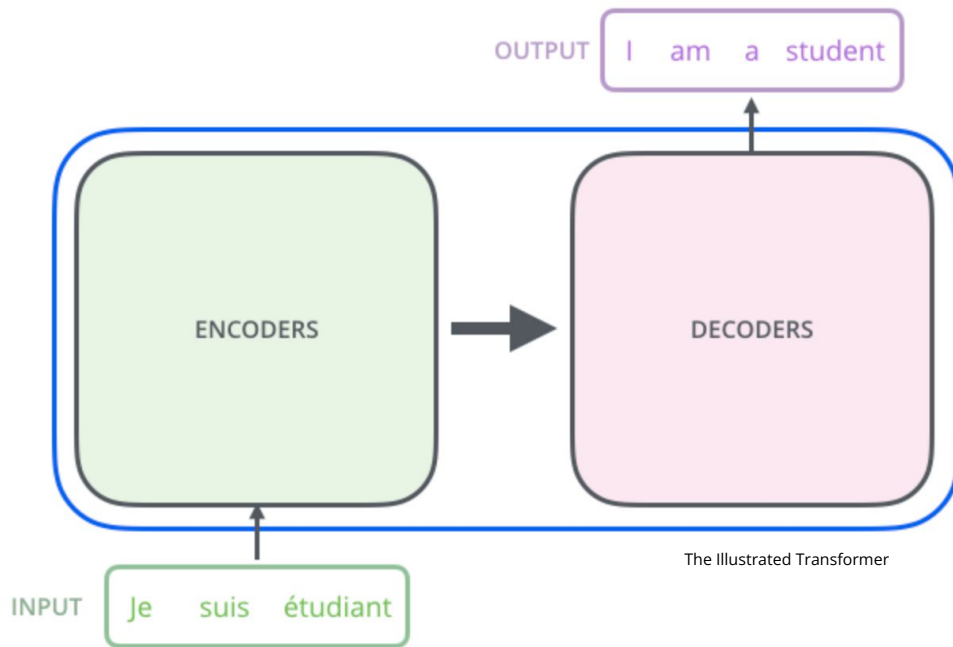
Transformers & Attention Mechanism:

Transformers - new architecture, which relies entirely on self-attention mechanisms. This enables the model to capture relationships in the data regardless of the distance between elements.

She poured water from the pitcher to **the cup** until **it was full**.

She poured water from **the pitcher** to the cup until **it was empty**.

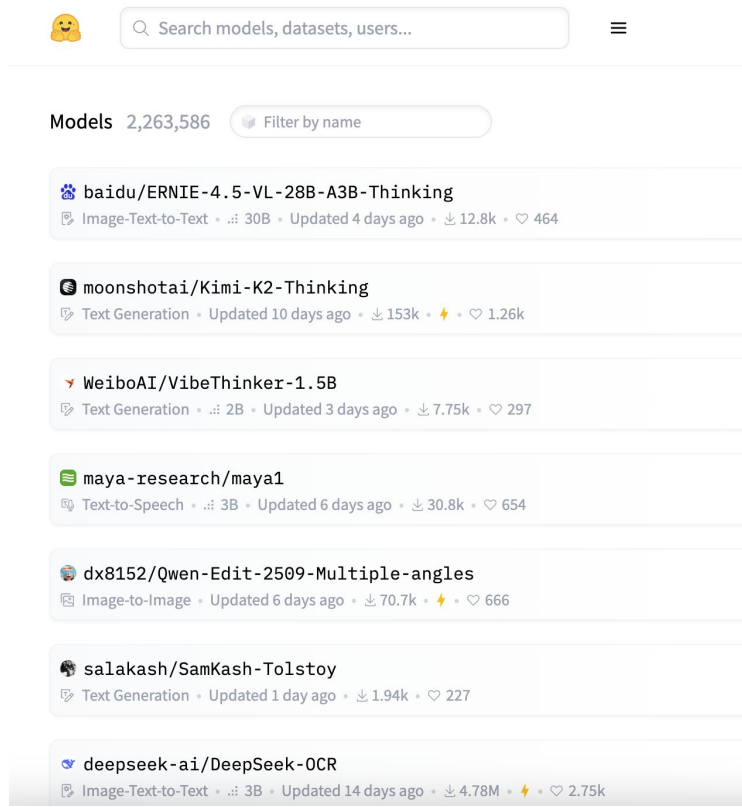
Transformer architecture



Encoder (left): The encoder receives an input and builds a representation of it (its features). This means that the model is optimized to acquire understanding from the input.

Decoder (right): The decoder uses the encoder's representation (features) along with other inputs to generate a target sequence. This means that the model is optimized for generating outputs.

Which model to choose?



The screenshot shows the Hugging Face Models page. At the top, there is a search bar with the placeholder text "Search models, datasets, users..." and a hamburger menu icon to its right. Below the search bar, the text "Models 2,263,586" is displayed, followed by a "Filter by name" button. A list of models is shown below, each with its name, icon, and some statistics:

- baidu/ERNIE-4.5-VL-28B-A3B-Thinking**
Image-Text-to-Text • 30B • Updated 4 days ago • 12.8k • 464
- moonshotai/Kimi-K2-Thinking**
Text Generation • Updated 10 days ago • 153k • 1.26k
- WeiboAI/VibeThinker-1.5B**
Text Generation • 2B • Updated 3 days ago • 7.75k • 297
- maya-research/maya1**
Text-to-Speech • 3B • Updated 6 days ago • 30.8k • 654
- dx8152/Qwen-Edit-2509-Multiple-angles**
Image-to-Image • Updated 6 days ago • 70.7k • 666
- salakash/SamKash-Tolstoy**
Text Generation • Updated 1 day ago • 1.94k • 227
- deepseek-ai/DeepSeek-OCR**
Image-Text-to-Text • 3B • Updated 14 days ago • 4.78M • 2.75k

2 263 586 models!

<https://huggingface.co/models>

Large Language Models. Best practices

Achieve best possible performance

- Use best models (could be LLM API)
- Use prompts with detailed task context, relevant information, instructions
- Experiment with prompt engineering techniques
- Experiment with few-shot examples that are 1) relevant to the test case, 2) diverse (if appropriate)
- Spend quality time optimizing a pipeline / "chain"

Optimize Cost

- Start optimizing for cost once you reach your expected performance
- Use techniques like Quantization, Distillation or smaller models trained on synthetic data

Prompt engineering



You don't need to be a data scientist or a machine learning engineer – everyone can write a prompt.

What is a prompt?

A **prompt** is natural language text that tells the AI what task to perform.

In a text-to-text language model, a prompt can take many forms:

- A question or query
- A command
- A longer statement with context, instructions, or past conversation

Aspects of the Prompt Engineering



Accessible to Everyone

You don't need to be a data scientist or ML engineer - anyone can write effective prompts.



Iterative Process

Crafting effective prompts requires experimentation and refinement.



Multiple Factors

Model choice, training data, configurations, word choice, and context all impact prompt effectiveness.

What Does Prompt Engineering Involve?

Carefully phrasing the query to guide the model's response:

- Specifying tone, style, or structure (e.g., formal vs. casual)
- Choosing the right words and grammar
- Providing relevant context or background
- Describing a persona or role for the AI to adopt

➡ The goal is to get more **accurate**, **reliable**, and **context-aware** outputs from the model.

Temperature Control



Low Temperature (0-0.3)

Deterministic, consistent outputs



Medium Temperature (0.4-0.7)

Balanced creativity and coherence



High Temperature (0.8-1.0)

More random, diverse outputs

Prompting techniques

Prompt engineering. Tips

<https://www.promptingguide.ai>

1. Start Simple
2. The Instruction:

```
### Instruction ###  
Translate the text below to Spanish:  
Text: "hello!"
```

3. Specificity: Be very specific about the instruction and task you want the model to perform
4. Avoid Impreciseness

Use 2-3 sentences to explain the concept of prompt engineering to a high school student.

Zero-Shot Prompting

<https://www.promptingguide.ai>

Prompt:

Classify the text into neutral, negative or positive.

Text: I think the vacation is okay.

Sentiment:

Output:

Neutral

Few-Shot Prompting

<https://www.promptingguide.ai>

Prompt:

A "whatpu" is a small, furry animal native to Tanzania. An example of a sentence that uses the word whatpu is:

We were traveling in Africa and we saw these very cute whatpus.

To do a "farduddle" means to jump up and down really fast. An example of a sentence that uses the word farduddle is:

Output:

When we won the game, we all started to farduddle in celebration.

[Brown et al. 2020](#)

System, contextual and role prompting

- **System prompting** sets the overall context and purpose for the language model. It defines the 'big picture' of what the model should be doing, like translating a language, classifying a review etc.
- **Contextual prompting** provides specific details or background information relevant to the current conversation or task. It helps the model to understand the nuances of what's being asked and tailor the response accordingly.
- **Role prompting** assigns a specific character or identity for the language model to adopt. This helps the model generate responses that are consistent with the assigned role and its associated knowledge and behavior.

Chain-of-Thought (CoT) Prompting

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

Prompt-Chaining

Prompt-Chaining: Decomposes a task into sequential subtasks, where each step builds on previous results

ReAct (reason & act)

Prompt:

Use Reasoning and Acting (ReAct) to solve this problem:

- **"Thought:** Describe your reasoning."
- **"Action:** (Options: Search, Do Nothing)"
- **"Observation:** Report the result of the action."

Repeat until you can answer.

Problem: *What is the capital of Australia, and what's the current temperature there?*

ReAct (reason & act). Output

Thought: I know that the capital of Australia is Canberra, but I need to search for the current temperature.

Action: Search "current temperature in Canberra."

Observation: The current temperature in Canberra is 18°C.

Thought: I have both pieces of information now.

Final Answer: The capital of Australia is Canberra, and the current temperature there is 18°C.

Automatic Prompt Engineering

Step1. Write the prompt which will generate the output variants.

Prompt	We have a band merchandise t-shirt webshop, and to train a chatbot we need various ways to order: "One Metallica t-shirt size S". Generate 10 variants, with the same semantics but keep the same meaning.
Output	1. I'd like to purchase a Metallica t-shirt in size small. 2. Can I order a small-sized Metallica t-shirt? 3. I'd like to buy one Metallica t-shirt in small. 4. One Metallica shirt, size small, please. 5. I want one Metallica t-shirt in size small. 6. I'd like to place an order for one Metallica t-shirt in small. 7. Small Metallica t-shirt, one please. 8. I'm looking for one Metallica t-shirt in small. 9. Can you get me one Metallica t-shirt in small? 10. One Metallica small t-shirt, please.

Automatic Prompt Engineering

Step 2. Evaluation

Evaluate all of the instruction candidates by scoring the candidates based on a chosen metric: BLEU or ROUGE.

BLEU (Bilingual Evaluation Understudy):

Measures precision: How much of the generated text matches the reference.

ROUGE (Recall-Oriented Understudy for Gisting Evaluation):

Focuses on overlap between reference and generated text.

How much of the reference content appears in the generated output.

Hands-on Prompt engineering
